Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/30/2009
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Simulated Microgravity and Radiation-Induced Bone Degeneration: Oxidative Stress- and p53-Dependent Mechanisms		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	10/01/2005	**End Date:**	09/30/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2010 per A. Chu/ARC (5/2010) NOTE: Received NCE to 3/31/2010 per A. Chu/ARC (7/2009) NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	No PI changes. Recently postdoctoral fellow, Dr. Kenji Yumoto accepted and started a new position elsewhere. Research continues with trained technical staff.		
COI Name (Institution):	Almeida, Eduardo A.C. (NASA Ames Research Center) Limoli, Charles (University of California Irvine) Vercoutere, Wenonah (NASA Ames Research Center) Searby, Nancy (NASA Ames Research Center)		
Grant/Contract No.:	INTRAMURAL		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	To determine the risks and mechanisms of tissue degeneration in the spaceflight environment, we will examine the combined effects of radiation and simulated weightlessness on bone. We will focus our studies on two important pathways, p53 and oxidative and nitrative stress-related species, in bone marrow progenitor populations because there is abundant evidence that the biological effects of weightlessness and those of radiation share these mechanism. Experiments will be conducted using the rodent hindlimb unloading model, the best validated animal model for simulating weightlessness, followed by irradiation. In Aim 1, we will test the hypothesis that simulated weightlessness of mice modulates the response of bone and bone marrow to radiation, both acutely and during tissue regeneration. In Aim 2 we will identify the target marrow-derived stem, progenitor and differentiated cell populations sensitive to the combined stimuli of simulated weightlessness and radiation. In Aim 3, we will explore the relevance of specific ROS/RNS and p53-dependent mechanisms for mediating cellular responses to combined simulated weightlessness and radiation. The results of these studies are expected to provide important information for the development of countermeasures both to help prevent degeneration of connective tissues such as bone during spaceflight and to improve subsequent tissue regeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project addresses several of the risks identified in NASA's Bioastronautics Critical Path Roadmap (BCPR) caused by long duration spaceflight. We will obtain basic new knowledge about bone tissue degeneration caused by musculoskeletal disuse and radiation at the molecular, cell, and tissue level, relevant to Earth, as well as in space. Results will also lead to the formulation of new countermeasures concepts and risk assessment strategies for bone and connective tissue degenerative diseases.
Task Progress:	During this reporting period (Year 4), we performed two sets of experiments at NASA's Space Radiation Laboratory at Brookhaven National Laboratory (using Fe-56), and have nearly completed sample processing on the latter, long duration experiment. The goals of the first set of experiments (NSRL 08A Run) are as follows: 1. Determine if simulated weightlessness worsens the effects of space relevant radiation. 2. Define the dose-dependence and/or threshold for radiation-induced loss of spongy (cancellous) bone. 3. Determine which type of bone cells are affected by radiation. The goals of the second set of experiments (NSRL 08C Run) were as follows: 1. Determine if prior hindlimb unloading of mice causes a shift in radiation-induced inhibition of differentiation of bone forming cells from marrow cells. 2. Identify the long term (6 months) consequences of exposure to high LET radiation during transient disuse. (Sample analysis is still in progress on this experiment). As expected on the basis of previous findings, disuse or irradiation each caused cancellous bone loss and combined, there were no greater effects of the two treatments in short term experiments. Despite the lack of evidence for combined effects on bone structure, analyses of whole bone mechanical properties showed that the mechanical properties of the spine appeared to be compromised by the combination of disuse and irradiation. We previously found that the maturation of bone-building cells, osteoblasts, from bone marrow cells was impaired by disuse and irradiation in the short term. We confirmed and extended these observations in a second experiment, and showed that disuse can make bone marrow more sensitive to radiation. These results led us to suspect stem and/or precursors of bone forming cells within bone marrow were damaged by combined treatment. These events would be expected to lead to a failure of the skeleton to recover from the negative effects of disuse. However, this did not appear to be the case in the short term. So we performed a long duration study, designed to uncover persistent changes bone due to disuse and radiation, if in fact they do occur in the course of progressive aging. Analyses of results from this experiment are still in progress. Although the magnitudes of differences are relatively small, the results do indicate that disuse and radiation together may worsen bone loss that occurs during natural aging. This study has particular relevance to skeletal recovery of astronauts after long duration spaceflight missions that may take place in the future, particularly outside Earth's orbit. What are the cellular mechanisms underlying these tissue-level changes? So far, we have evidence that both cells within the lineage of bone-forming osteoblasts and the lineage of bone-resorbing osteoclasts play a role in these degenerative changes in bone.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Globus RK, Limoli CL, Yumoto K, Searby ND, Almeida EAC. "Influence of total body irradiation with 56Fe at low doses on skeletal microarchitecture and bone cell function in adult mice." Presented at DOE Low Dose Radiation Investigator Workshop, Bethesda, Maryland, April 2009. DOE Low Dose Radiation Investigator Workshop, Bethesda, Maryland, April 2009. , Apr-2009
Abstracts for Journals and Proceedings	Globus RK, Limoli CL, Almeida EAC, Searby ND. "Heavy particle radiation and simulated weightlessness adversely affect remodeling bone." NASA Radiation Program PI Workshop/Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 2009. NASA Radiation Program PI Workshop/Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 2009. , Jul-2009
Articles in Peer-reviewed Journals	Kondo H, Searby ND, Mojarrab R, Phillips J, Alwood J, Yumoto K, Almeida EA, Limoli CL, Globus RK. "Total-body irradiation of postpubertal mice with (137)Cs acutely compromises the microarchitecture of cancellous bone and increases osteoclasts." Radiation Research 2009 Mar;171(3):283-9. PMID: 19267555 , Mar-2009
Dissertations and Theses	Alwood JA. "Radiation and mechanical unloading effects on mouse vertebral bone: Ground based models of the spaceflight environment." Doctor of Philosophy award, Department of Aeronautics and Astronautics, Stanford University, pending final acceptance in September, 2009. , Sep-2009